PROBLEM VF6

Oil Rising in a Vertical Tube

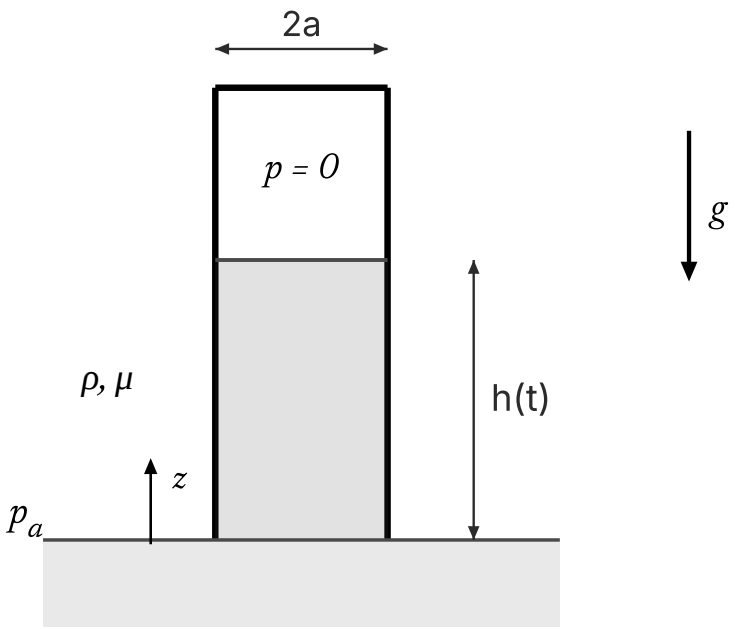
Original statement · VF6.pdf

Quasi-steady Poiseuille flow | Rising column | Force on the pipe

OFFICIAL PROBLEM STATEMENT

The bottom end of an empty vertical tube of radius a, closed at the top, is put in contact with a pool of oil. Because of the ambient overpressure, the liquid begins to flow into the tube, forming a column of increasing height $h(t)$ whose evolution in time is to be investigated assuming that the motion is dominated by viscosity. In particular:

- Obtain the value of the height h_∞ corresponding to the equilibrium position, reached asymptotically for large times.
- Give the condition that determines whether the motion is dominated by viscosity.
- Obtain the evolution of $h(t)$ as well as the pressure distribution along the pipe $p(x,t)$ for $0 < x < h$.
- Compute the force acting on the pipe as a function of time $\bar{F} = F_z(t)\bar{e}_z$.



Statement figure (redrawn from VF6.pdf): a vertical tube of radius a, closed at the top, with its lower end in a pool at p_a; the oil column of height h(t) has p = 0 above it; z is the vertical axis from the pool

Source: VF6.pdf (page 1, statement and handwritten key; no date printed). Notes.pdf, Chapter 7: steady flow in a circular pipe, Eqs. 7.13-7.16; slender flows, Eqs. 7.40-7.41; pipe flow, Eqs. 7.47-7.51. The sheet answers the four bullets in the margin order 1) to 4).

Worked solution

Hypotheses, Data and Unknowns

- The tube is evacuated above the liquid (the figure marks $p = 0$ there), so the free surface of the column is at pressure zero; the pool is at the atmospheric pressure p_a at the level of the lower end, which is the origin of the vertical axis z (the sheet writes $x \equiv z$ for the axial coordinate).
- The liquid is incompressible with constant ρ and μ; the column is slender, $h \gg a$ for most of the process.
- Reduced pressure: $P = p + \rho g z$.

- The atmosphere also acts on the outer face of the closed top end of the tube (this is what the sheet's force balance includes).

Symbol	Meaning	SI unit
a	radius of the tube	m
ρ, μ	density, viscosity of the oil	kg/m^3, kg/(m s)
p_a	ambient pressure	Pa
$h(t)$, h_∞	height of the column and its final value	m

Equilibrium Height

When $t \to \infty$ the liquid is at rest and the reduced pressure is uniform, $P = p + \rho g z =$ constant. At the bottom ($z = 0$) it equals p_a and at the free surface ($z = h_\infty, p = 0$) it equals $\rho g h_\infty$:

$$p_a = \rho g h_\infty \implies h_\infty = \frac{p_a}{\rho g}$$

Criterion for Viscous Dominance

The viscous force dominates the convective acceleration (and the local acceleration, since $t_0 \sim L/V_L$) when, Eq. (7.41),

$$\frac{\rho V_L a}{\mu}\,\frac{a}{L} \ll 1$$

with the characteristic length $L \sim h_\infty$ and the velocity estimated by assuming that viscosity dominates, Eq. (7.44), $V_L \sim \frac{a^2}{L}\frac{p_a}{\mu}$ (the pressure difference that drives the flow is p_a):

$$\frac{\rho a^4 p_a}{\mu^2 h_\infty^2} = \frac{\rho a^4 p_a}{\mu^2}\,\frac{\rho^2 g^2}{p_a^2} = \frac{\rho^3 g^2 a^4}{\mu^2 p_a} \ll 1$$

(the key writes the characteristic transverse length as D; here it is the radius a, which only changes a numerical factor).

Evolution of the Column

Poiseuille flow in the column (Eqs. 7.13-7.15), with $P_l = -\partial P/\partial z$ uniform along the column at each instant:

$$v_z = \frac{P_l}{4\mu}\left(a^2 - r^2\right), \qquad Q = \frac{\pi a^4}{8\mu} P_l$$

The reduced pressure falls from $P(0) = p_a$ to $P(h) = \rho g h$ over the length h, so $P_l = (p_a - \rho g h)/h$. The mean velocity of the column is the velocity of its free surface, $Q = \pi a^2\, dh/dt$ (the column grows, so the flux into the tube is positive upward; the key writes $-\pi a^2 dh/dt$ and corrects the sign by hand). Hence

$$\frac{dh}{dt} = \frac{a^2}{8\mu}\,\frac{p_a - \rho g h}{h} = \frac{a^2 p_a}{8\mu}\,\frac{1 - h/h_\infty}{h}$$

Separating variables, with $h(0) = 0$ and $u = h/h_\infty$ ($h\,dh = h_\infty^2\, u\,du$):

$$\int_0^h \frac{h'\,dh'}{1 - h'/h_\infty} = h_\infty^2 \int_0^{h/h_\infty} \frac{u\,du}{1-u} = \frac{a^2 p_a}{8\mu}\,t, \qquad \int \frac{u\,du}{1-u} = \int \left(-1 + \frac{1}{1-u}\right) du = -u - \ln(1-u)$$

$$\frac{a^2 p_a}{8\mu h_\infty^2}\,t = \ln\frac{1}{1 - h/h_\infty} - \frac{h}{h_\infty}$$

Discrepancy with the official solution

The key writes this relation, and the equation of dh/dt before it, without the factor a^2: $\frac{P_a}{8\mu h_\infty^2}t = \ln\frac{1}{1-h/h_\infty} - \frac{h}{h_\infty}$. That cannot be right dimensionally: $p_a t/\mu$ is dimensionless, so the left side has the dimensions L^{-2} while the right side is dimensionless; and $dh/dt = (p_a/8\mu)(1-h/h_\infty)/h$ has the dimensions $T^{-1}L^{-1}$ instead of LT^{-1}. The factor a^2 comes from $Q = \pi a^2 dh/dt$ and $Q = \pi a^4 P_l/(8\mu)$, whose ratio is a^2 (the key strikes out the πa^2 in that step). The same slip appears in the pressure-gradient lines of the key, written $8\mu Q/(\pi a^2)$, where the Poiseuille flux gives $8\mu Q/(\pi a^4)$. The results above keep the factor.

As $t \to \infty$, $\ln\frac{1}{1-h/h_\infty} \to \infty$ and $h \to h_\infty$. For short times $\ln\frac{1}{1-u} - u \simeq \frac{u^2}{2}$, so $h \simeq a\sqrt{p_a t/(4\mu)}$: the column grows as $\sqrt{t}$ (the laminar, slender-flow law), independent of gravity.

Pressure Distribution

The reduced pressure is linear, $P = p_a - P_l z$, with $P_l = (p_a - \rho g h)/h$. The actual pressure $p = P - \rho g z$:

$$p(z,t) = p_a - \frac{p_a - \rho g h}{h}\, z - \rho g z = p_a - \frac{p_a}{h}\, z = p_a\left(1 - \frac{z}{h(t)}\right), \qquad 0 < z < h$$

linear from p_a at the pool to zero at the top of the column; the weight terms cancel (the key writes the equivalent $P + \rho g z = p_a - \frac{8\mu Q}{\pi a^4}x$, boxed, with the same slip in the power of a).

Force on the Pipe

The liquid acts on the pipe through the shear stress of the wall. With the normal from the wall into the liquid, $\bar{n} = -\bar{e}_r$, $\tau_p = -\mu\,(\partial v_z/\partial r)_{r=a} = P_l a/2$ (Eq. 7.16), along $+\bar{e}_z$ (the rising liquid drags the wall upward). Over the lateral area $2\pi a h$:

$$F_{visc} = 2\pi a h \cdot \frac{a}{2}\,\frac{p_a - \rho g h}{h} = \pi a^2\,(p_a - \rho g h)$$

The pressure inside the tube above the liquid is zero, and the atmosphere acts on the outer face of the closed top end with the force $-p_a \pi a^2 \bar{e}_z$ (the closed end is a disc of area πa^2). The total force:

$$\bar{F} = \pi a^2 \left(p_a - \rho g h\right) \bar{e}_z - p_a \pi a^2 \bar{e}_z = -\rho g h \, \pi a^2 \, \bar{e}_z$$

i.e. $F_z(t) = -\rho g \pi a^2 h(t)$: the weight of the column (downward), which is a good check: the force on the pipe equals the weight of the liquid it holds, since the liquid is quasi-static in the vertical momentum balance of the whole column ($\pi a^2 p_a$ at the bottom, minus the weight, equals the shear transmitted to the wall).

Answers and Checks

- Equilibrium height: $h_\infty = p_a/(\rho g)$.
- Viscous dominance: $\rho^3 g^2 a^4/(\mu^2 p_a) \ll 1$.
- Height: $\frac{a^2 p_a}{8\mu h_\infty^2} t = \ln \frac{1}{1-h/h_\infty} - \frac{h}{h_\infty}$; pressure: $p = p_a(1 - z/h)$.
- Force on the pipe: $\bar{F} = -\rho g \pi a^2 h(t) \, \bar{e}_z$.
- Dimensions: $a^2 p_a t/(\mu h_\infty^2)$ is $L^2 \cdot ML^{-1}T^{-2} \cdot T/(ML^{-1}T^{-1}L^2) = 1$.
- Limits: at $t \to 0$, $h \to a\sqrt{p_a t/(4\mu)}$; at $t \to \infty$, $h \to h_\infty$ and $F_z \to -\pi a^2 p_a$ (the atmosphere on the top end, with the shear zero).
- Check against the official key: VF6.pdf gives $h_\infty = p_a/(\rho g)$, the criterion $\rho^3 g^2 D^4/(\mu^2 p_a) \ll 1$, the integral for $h(t)$ and $\bar{F} = -\rho g h \pi a^2 \bar{e}_z$; the key's $h(t)$ relation lacks the factor a^2, as noted above.